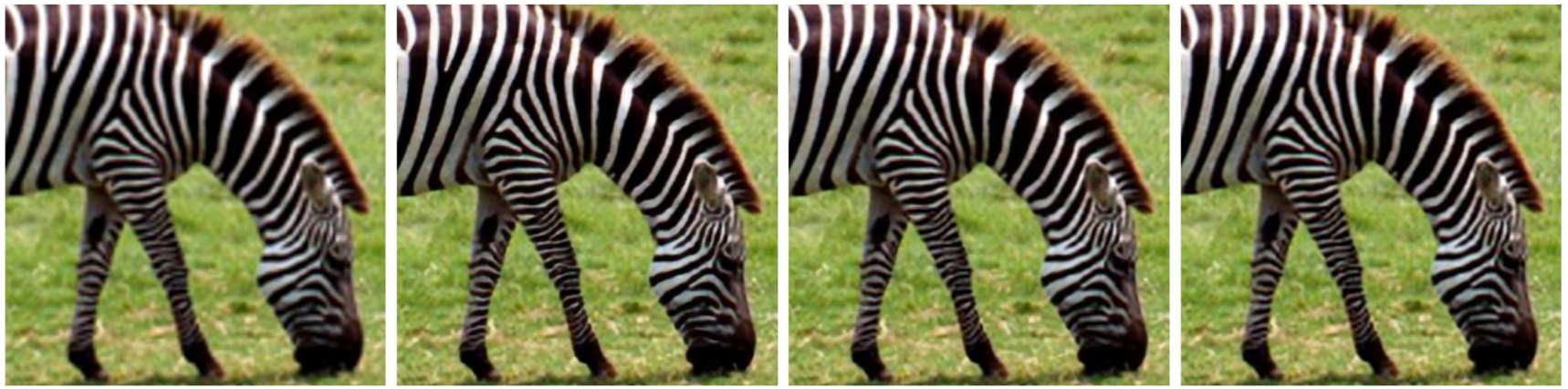
Modern methods



(a) Bicubic (b) SRCNN (c) A+ (d) RAISR



(e) Bicubic (f) SRCNN (g) A+ (h) RAISR

From Romano, et al: RAISR: Rapid and Accurate Image Super Resolution, https://arxiv.org/abs/1606.01299

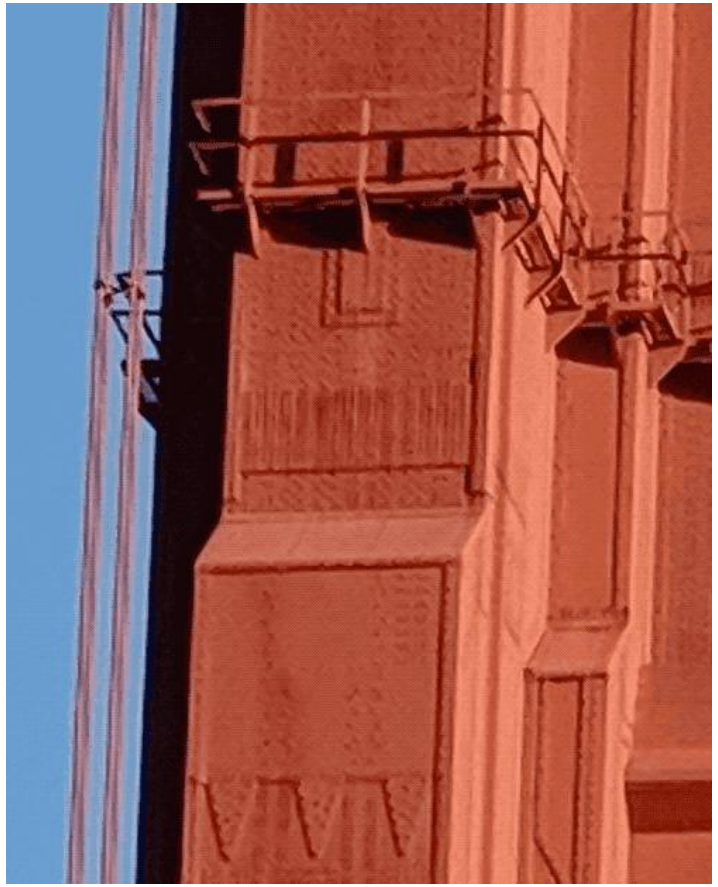
Super-resolution with multiple images

- Can do better upsampling if you have multiple images of the scene taken with small (subpixel) shifts
- Some cellphone cameras (like the Google Pixel line) capture a **burst** of photos
- Can we use that burst for upsampling?

Google Pixel 3 Super Res Zoom



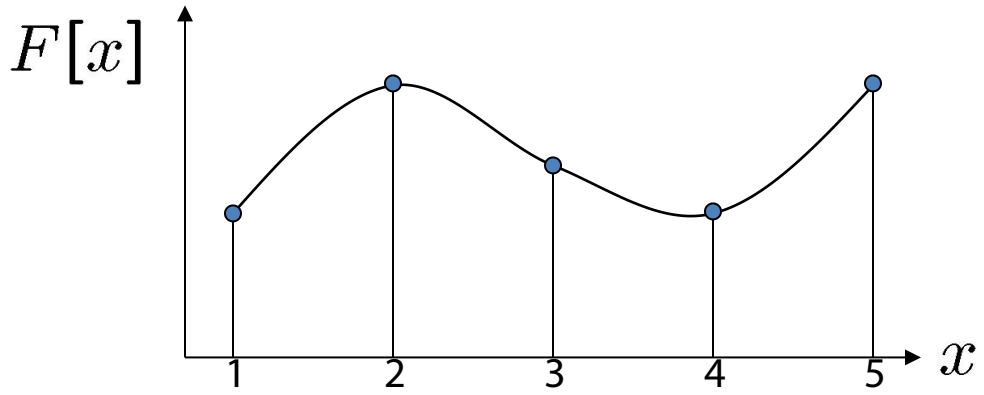
Effect of hand tremor as seen in a cropped burst of photos, after global alignment



Example photo with and without super res zoom (smart burst align and merge)

https://ai.googleblog.com/2018/10/see-better-and-further-with-super-res.html

Image interpolation



d = 1 in this example

Recall that a digital images is formed as follows:

$$F[x, y] = \text{quantize}\{f(xd, yd)\}$$

- It is a discrete point-sampling of a continuous function
- If we could somehow reconstruct the original function, any new image could be generated, at any resolution and scale

Summary

- Key points:
  - **Subsampling an image** can cause aliasing. Better is to blur (“pre-filter”) to remove high frequencies then downsample
  - If you repeatedly blur and downsample by 2x, you get a Gaussian pyramid
  - **Upsampling an image** requires interpolation. This can be posed as convolution with a “reconstruction kernel”

Questions?